Hera Choi

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Education

University College London — London, UK
BSc Computer Science — *June 2028*

- Average grade: **77%** (First Class)
- Relevant modules: Systems Engineering, Software Engineering, OOP with Java, Algorithms with Python

Magdalen College School — Oxford, UK
A-levels: Maths (A), Computer Science (A*), Physics (A*), Further Maths (A)* — *Sept. 2023 – July 2025*

Competitions

PACT: Flat-share Hub (IKU Womxn in STEM Hackathon) — Oct 2026

- Decided project scope and task assignment with the team, cutting the flatmate finder feature for a focused idea
- Set up the local environment for non-technical teammates to continue development despite Lovable credit limits
- Implemented the email nudge feature with Copilot using Resend, managing DNS records on Porkbun and Vercel
- Achieved 1st place out of 22 teams on the Wise track

Cambridge Battlecode Finals (sponsored by Jump Trading) — April. 2026 - May 2026

- Profiled bot execution in local test runs using cProfile and SnakeViz to identify a hot path; optimised execution time to a third, improving game resource pathfinding within a 2000ms compute budget
- Built an A/B evaluation pipeline with Claude Code to benchmark strategy variants against the team baseline in overnight simulations; statistically evaluated 30+ changes across 8 bot versions
- Achieved joint 10th place against 500+ teams

National Cipher Challenge: Northrop Grumman Prize | *Pandas, scikit-learn* — Sept. 2024 – Dec. 2024

- Built a ciphertext fingerprinting toolkit from index-of-coincidence and n-gram statistics to guess the cipher family, reduced search space of the team's 33 solvers to apply in weekly challenges
- Extended the toolkit into a scikit-learn classifier, cleaning 24 years of past challenges to build the dataset
- Engineered features by merging statistically indistinguishable cipher labels and compared models to raise classifier accuracy to 92% with Random Forest
- Achieved 8th out of 7,000+ teams - top 0.1% nationally

Projects

Full-stack inventory web app | *Python, Flask, React, PostgreSQL, Supabase* — Aug. 2024 – May 2025

- Built a system to address manual searches through prop storage and prop allocation conflicts across productions
- Conducted interviews and surveys with crew members to gather requirements and iterate on MVP objectives
- Designed a PostgreSQL schema using many-to-many relations and inheritance to model props and productions
- Built a Flask REST API for CRUD operations with a React frontend for rapid prototyping
- Performed black-box and user testing; documented design and evaluation in a 124-page report

Experience

Programming Mentor — Oct. 2026 – Present
University College London — *London*

- Mentored 10 1st-year students with diverse programming experience in using C, Haskell, Java and Python through weekly sessions supporting coursework and projects

Freelance Vocalist/Songwriter — Nov. 2022 – Present
Self-employed — *Remote*

- Delivered 70+ commissions for clients from 12 countries (4.9 average rating); 7 regular clients
- Communicated project scope and details with clients with various levels of musical proficiency

Technical Skills

Languages: Python, C++, C, Java, JavaScript, SQL, Haskell
Frameworks & Libraries: Flask, React, Material UI, scikit-learn, Matplotlib
Tools & Systems: Linux, Git, PostgreSQL, MQTT